

Joseph E. Velasquez

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Education

California State University Fullerton

B.S. Computer Science

Riverside City College

A.S. Computer Science

Fullerton, CA

Aug 2025

Riverside, CA

May 2022

Experience

Inland Neurosurgery Institute - Software and Systems Engineer

Aug 2025 – Oct 2025

- Supported integration and deployment of a new Electronic Medical Records (EMR) system, ensuring minimal disruption to clinical operations
- Resolved hardware, software, and network issues during EMR migration to ensure continuity of clinical operations
- Built and maintained automation workflows using Microsoft Power Automate, reducing manual administrative workload and increasing efficiency
- Assisted in maintaining cybersecurity measures, including system monitoring, patching, and basic threat mitigation
- Developed a patient scheduling chatbot using local LLM, enabling automated appointment booking and reducing administrative workload

Residential IT Support Technician (ResNet) - University of California, Santa Barbara (UCSB)

Aug 2018 – June 2019

- Provided Tier 1 technical support for hardware, software, and network connectivity issues in a residential campus environment.
- Diagnosed and resolved Windows, macOS, Wi-Fi, Ethernet, TCP/IP, DNS, and DHCP issues.

Projects

Homelab Cloud + AI Platform - Self-Hosted Kubernetes/GitOps Infrastructure

Sep 2026

- Deployed a two-node Kubernetes (K3s) cluster (Raspberry Pi 5 control plane + ARM64 VM worker) with GitOps via Argo CD, GitHub Actions CI/CD, SOPS-encrypted secrets, and Ansible/Terraform for configuration as code
- Found and fixed a production-style security gap where firewall rules never applied to Kubernetes traffic due to iptables chain ordering, verified with real blocked-vs-allowed traffic tests
- Built a FastAPI backend (rate limiting, retries, Prometheus metrics) with RAG over a local Ollama LLM (PostgreSQL/pgvector, Redis) and a streaming chat assistant that searches an offline Wikipedia archive
- Load tested with k6 (0% errors at ~480 req/s, cross-checked in Prometheus/Grafana); failure injection exposed a bug where a dead AI backend hung requests 180s+, fixed to fail cleanly in ~18s
- Automated encrypted backups with email alerting and a recovery rehearsal that restores the control plane into a throwaway VM, verifying the same cluster CA and 23 running pods

ParkingPulse - AI Vehicle Tracking & Data System

Aug 2024 – May 2025

- Engineered a real-time AI vehicle tracking platform using Python, OpenCV, and YOLOv8 for intelligent object detection and analytics.
- Designed and implemented RESTful APIs to handle real-time data ingestion, processing, and retrieval
- Integrated Firebase for authentication and cloud-based data storage, enabling scalable and secure data management
- Optimized system performance by reducing processing latency through frame skipping and model tuning
- Implemented data pipelines for continuous processing and storage of vehicle detection events

RAG Office Assistant - Local-LLM Retrieval-Augmented Chatbot

Sep 2025

- Built a retrieval-augmented generation (RAG) chatbot in FastAPI, using FAISS vector search and sentence-transformer embeddings to ground responses in a real knowledge base instead of relying on model hallucination
- Integrated Ollama for fully local LLM inference so patient data and queries never leave the machine, streaming responses to the browser over Server-Sent Events
- Added a relevance-threshold filter on retrieval so out-of-scope questions are refused instead of answered with fabricated context
- Identified and fixed a reflected XSS vulnerability in the chat interface by replacing unsafe HTML injection with safe DOM rendering
- Containerized the API and LLM services with Docker Compose and open-sourced the project on GitHub

Undergraduate Research Assistant — UAV Data & Vision Systems

Aug 2024 – Mar 2025

- Developed OpenCV-based computer vision pipelines for real-time object tracking in UAV simulations
- Processed geospatial routing data using OSMnx to improve autonomous navigation efficiency
- Simulated autonomous systems in Webots to evaluate navigation performance and system reliability

Skills & Interests

- Languages: Python, JavaScript, TypeScript, C++, SQL
- Backend: Node.js, FastAPI, Flask, REST APIs
- Frontend: SvelteKit, HTML/CSS
- Cloud & DevOps: AWS (EC2, S3), GCP, Docker, Kubernetes (K3s), Argo CD, Ansible, Terraform, CI/CD
- Databases: PostgreSQL (pgvector), Redis, MongoDB, Firebase Firestore
- AI / Machine Learning: OpenCV, YOLOv8, FAISS, sentence-transformers, RAG, Ollama, Real-Time Data Processing
- Tools: Git, Linux, Prometheus, Grafana, k6

Certifications

- AWS Knowledge: Serverless
- Google Cloud: App Dev Environment
- Google Cloud: Load Balancing